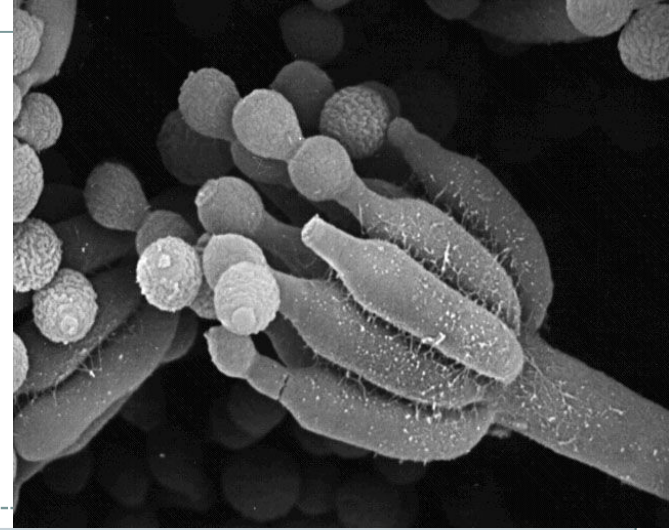




Microbes in soil

**P. NEELAMANIE YAPA
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RAJARATA UNIVERSITY OF
SRI LANKA**



Learning Outcomes are to:



- describe morphological indications of different soil microorganisms;
- identify the representatives of the main soil microorganism groups;
- understand the course of microbiological processes and know their main agents
- Identify activity and interactions of different microorganisms in soil
- use of microorganisms in improving soil health.

Soil



Soils correspond to the surface of the Earth's crust (lithosphere: litho = rock).

They develop as a result of the weathering of geological rock formations.



Soils Defined

Natural Body that Occurs on the land Surface

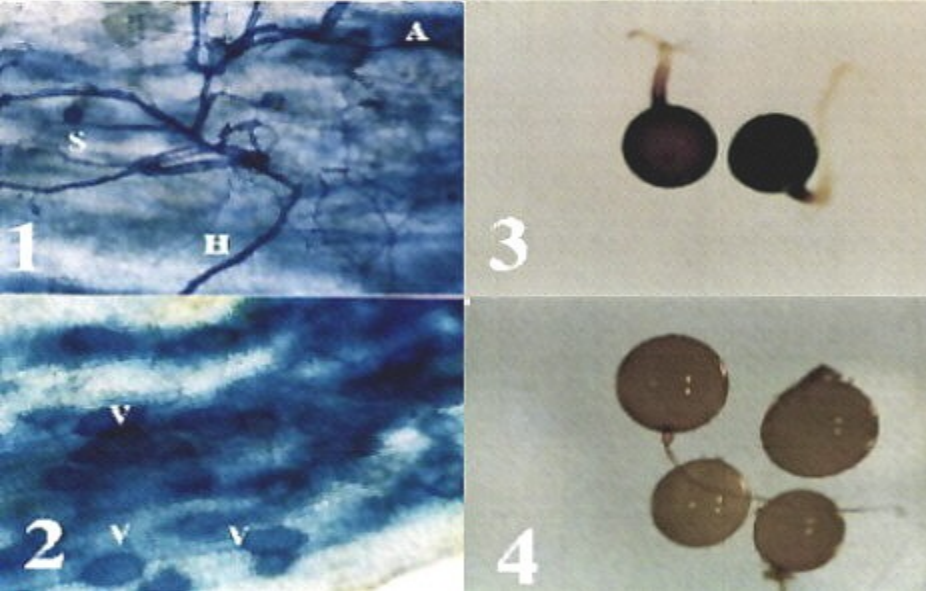
Consists of Distinct Horizons or Layers

The ability to support rooted plants in a natural environment

Upper Limit is Air or Shallow Water

Lower Limit is Bedrock or Limit of Biological Activity

Classification based on a typical depth of 2 m or approximately 6.0 feet

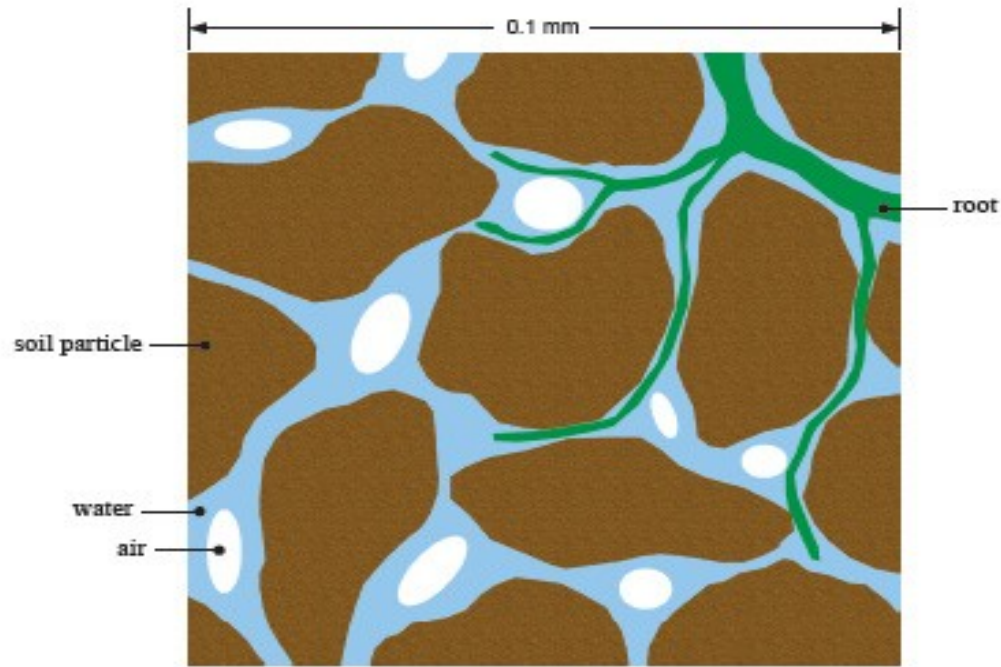
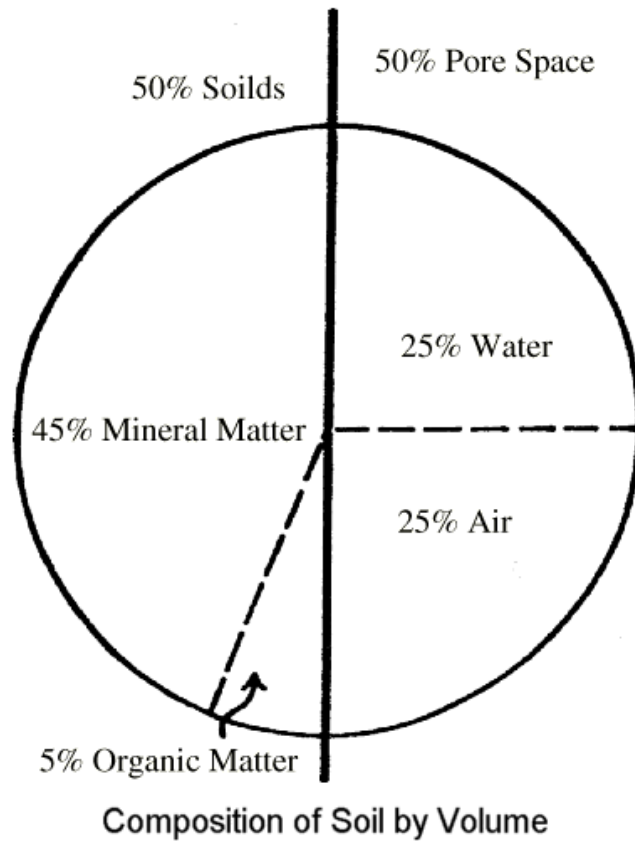


Soil microbiology

They are found in association with air (i.e. the soil atmosphere) and an aqueous phase (i.e. the soil solution) in the pores existing between particles and/or aggregates.



Soil Composition



Constantly changing
Mineral

rock derivative

Pore space

water

air

Organic Matter

plant and animal matter

living and dead



Five Soil Formation Factors

Organisms

Climate

Time

Topography and Landscape
Setting

Parent Material

**Point: Soils are Created in Geological Time, but can
be destroyed in a very short time**



Soil Organic Matter

Composition

Carbon Cycle

living and dead
plant and animal
dynamic system

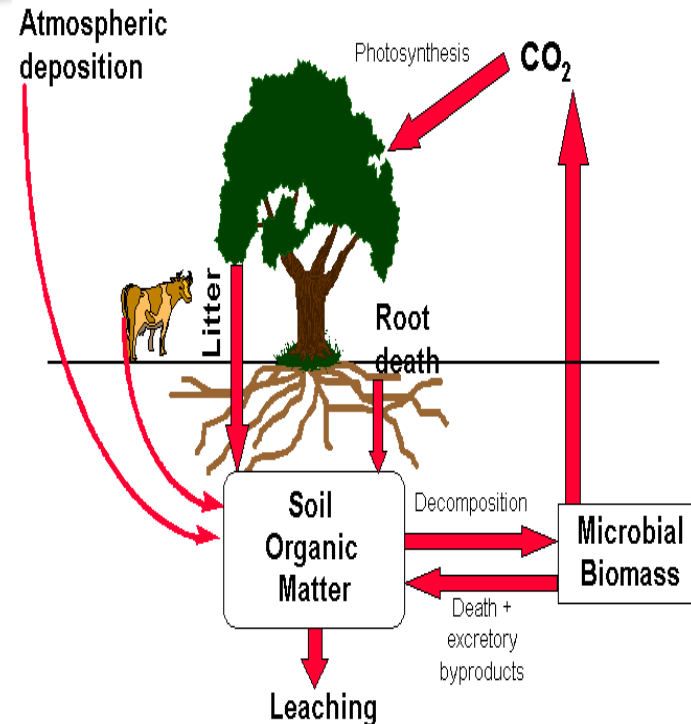
Advantages

Improves structure

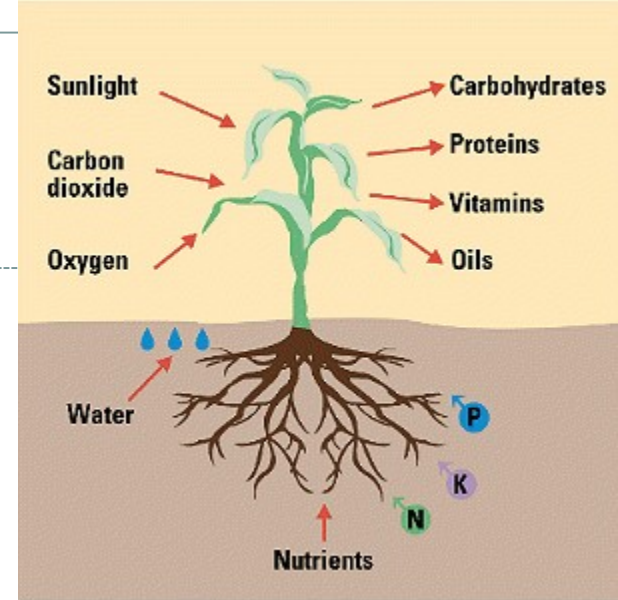
increased infiltration
increased water holding
capacity
increased workability

increased CEC

OM has high number of
exchange sites

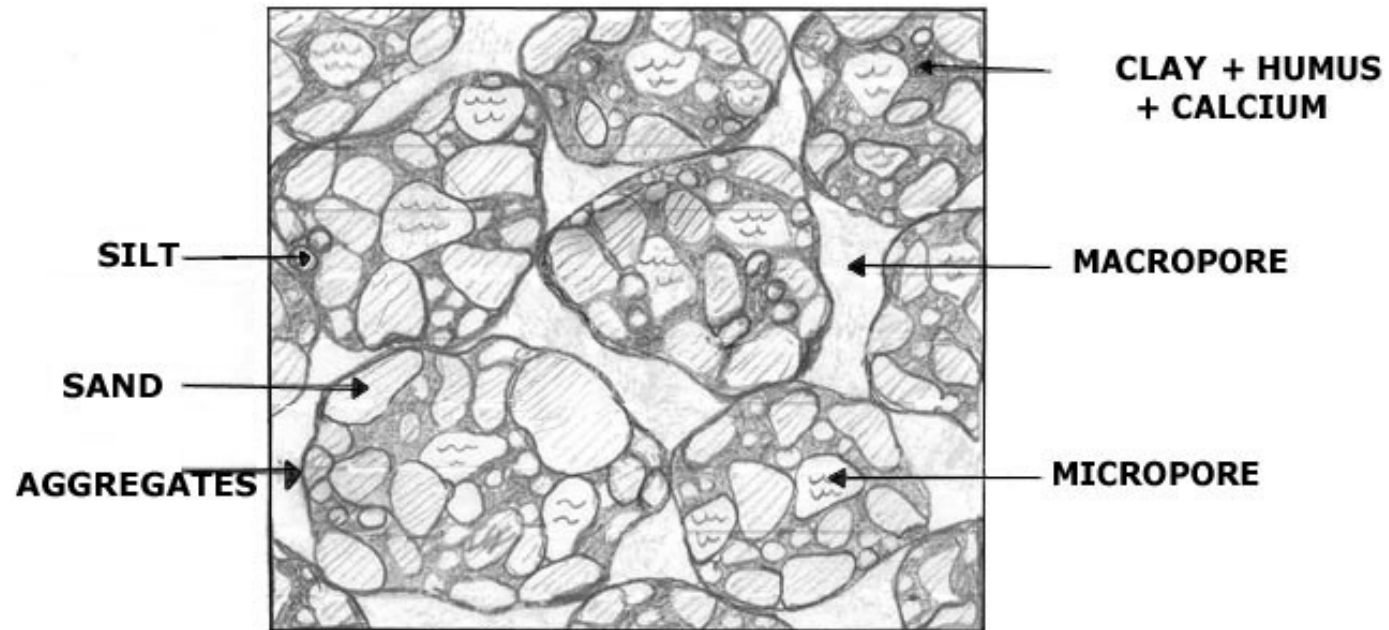


Chemical Properties



Soil pH
Cation Exchange Capacity
Organic matter content
Clay mineralogy
Essential Plant Nutrients
Nutrient Movement
Root/soil interaction

Types of Pores



Macropores (> 1,000 microns)- large

Mesopores (10 to 1,000 microns)- Medium

Micropores (< 10 microns)- Small

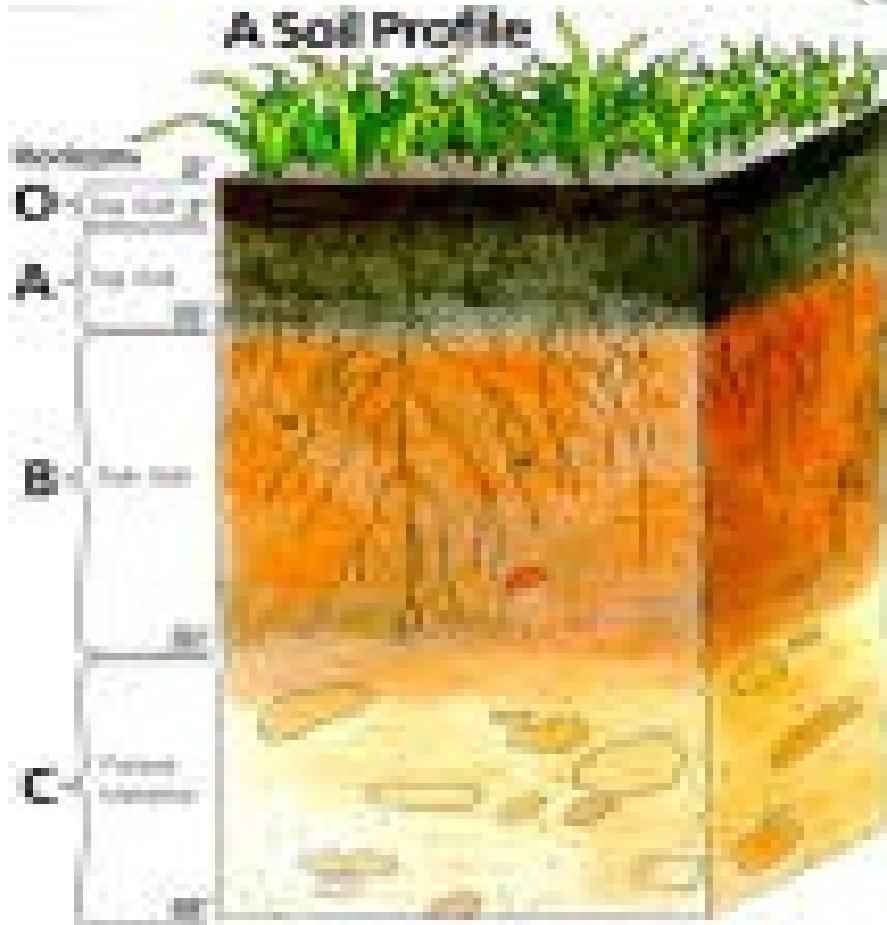
Water Stable Aggregates

Aggregates on left are more water stable, i.e., aggregate stays together and do not separate into its components,



Water Stable Aggregates

Soil Horizons



Layer of Soil
Parallel to Surface
Properties a
function of
climate,
landscape setting,
parent material,
biological activity,
and other soil
forming
processes.

Image Source: University of Texas, 2002

Horizons (A, E, B,
C, R, etc.)

Soil Horizons

O- Organic Horizons



Dark in Color Because of Humus Material
- 1,000,000 bacteria per cm³

Organic Layers of
Decaying Plant and
Animal Tissue

Aids Soil Structural
Development

Helps to Retain
Moisture

Enriches Soil with
Nutrients

Infiltration Capacity
function of Organic
Decomposition

Organic Matter
Critical in
Maintaining Water

Soil Horizons

A Horizons: “Topsoil”



Mineral Horizon
Near
Surface

Moves Humic and
Minerals from O
Horizon into A
horizon

Soil dark in color,
coarser in texture,
and high porosity

The Soil Food Web



First trophic level:
Photosynthesizers

Second trophic level:
Decomposing Mutualists
Pathogens, Parasites
Root-feeders

Third trophic level:
Shredders
Predators
Grazers

Fourth trophic level:
Higher level predators

Fifth & higher trophic level:
Higher level predators

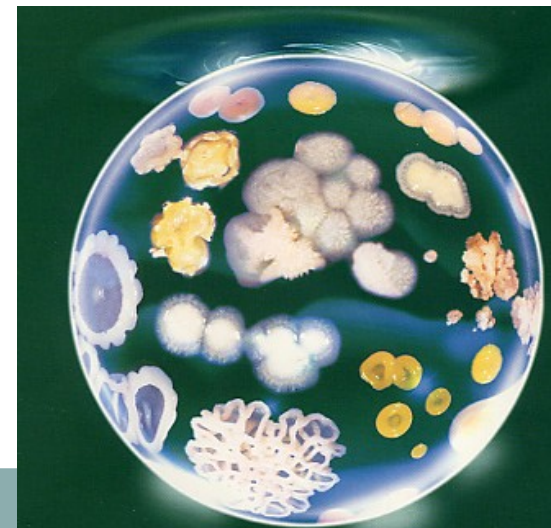


Soil - a favorable habitat for microorganisms inhabited by a wide range of microorganisms, including bacteria, fungi, algae, viruses and protozoa.

Microorganisms are found in large numbers in soil - usually between one and ten million microorganisms are present per gram of soil.



Bacteria and fungi being the most prevalent. However, the availability of nutrients is often limiting for microbial growth in soil and most soil microorganisms may not be physiologically active in the soil at a given time.



Microorganisms and their significance:



Almost every chemical transformation taking place in soil involves active contributions from soil microorganisms.

Active role in soil fertility as a result of their involvement in the cycle of nutrients like carbon and nitrogen, which are required for plant growth.

Microorganisms and their significance:



Responsible for the decomposition of the organic matter entering the soil and therefore in the recycling of nutrients in soil.

Certain soil microorganisms such as mycorrhizal fungi can also increase the availability of mineral nutrients (e.g. phosphorus) to plants.

Microorganisms and their significance:



Can increase the amount of nutrients present in the soil.

The microorganisms, which improve the fertility status of the soil and contribute to plant growth, have been termed 'biofertilizers'.

Microorganisms and their significance:



Soil microorganisms produce compounds (such as vitamins and plant hormones) that can improve plant health and contribute to higher crop yield.

‘Phytostimulators’ are currently studied for possible use as microbial inoculants to improve crop yield.

Microorganisms and their significance:



Some soil microorganisms are pathogenic to plants. Large numbers are routinely found in the soil and many of them can infect the plant through the roots.

Microorganisms and their significance:



Antagonistic to the pathogens and can prevent the infection of crop plants.

- Competition for nutrients and/or production of inhibitory compounds such as secondary metabolites (antimicrobial metabolites and antibiotics) and extracellular enzymes.

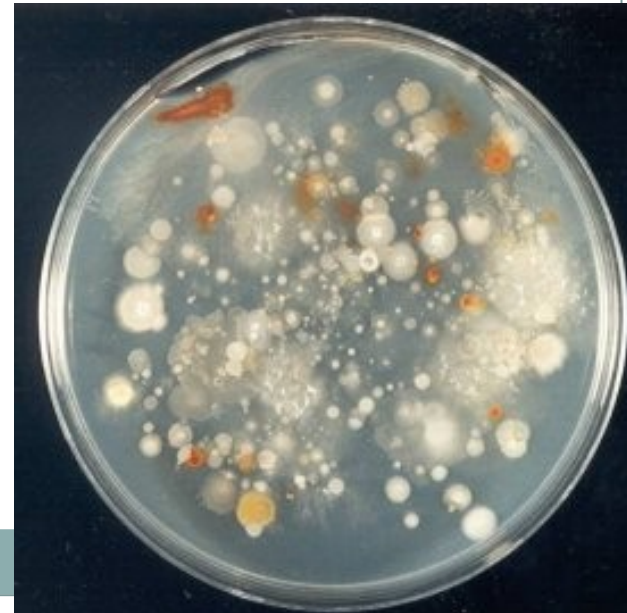
Major groups of soil microorganisms :



Bacteria:

Numerically abundant

Along with fungi, most important decomposers of organic matter



Major groups of soil microorganisms :

Bacteria:

Specialized groups participate in all biogeochemical cycles

Their extracellular polymers help bind soil particles into aggregates

Some form beneficial or pathogenic interactions with plants



Major groups of soil microorganisms :



Nitrogen fixing organisms: bacteria

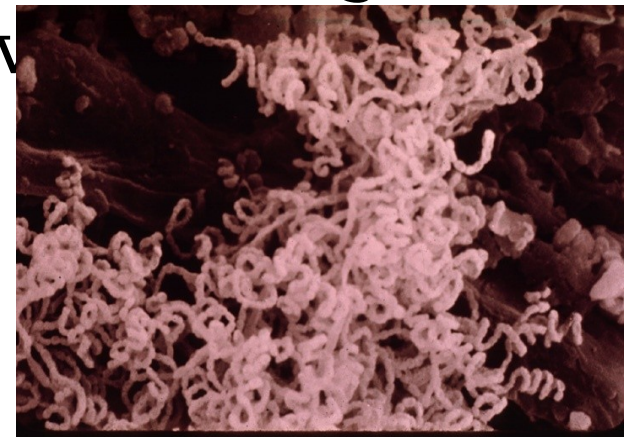


Major groups of soil microorganisms :



Actinomycetes:

- specialized filamentous prokaryotes
- participate in decomposition of complex organic compounds
- produce many 2nd metabolites, e.g., antibiotics, geosmins (earth odor) that give soil its characteristic distinctiv



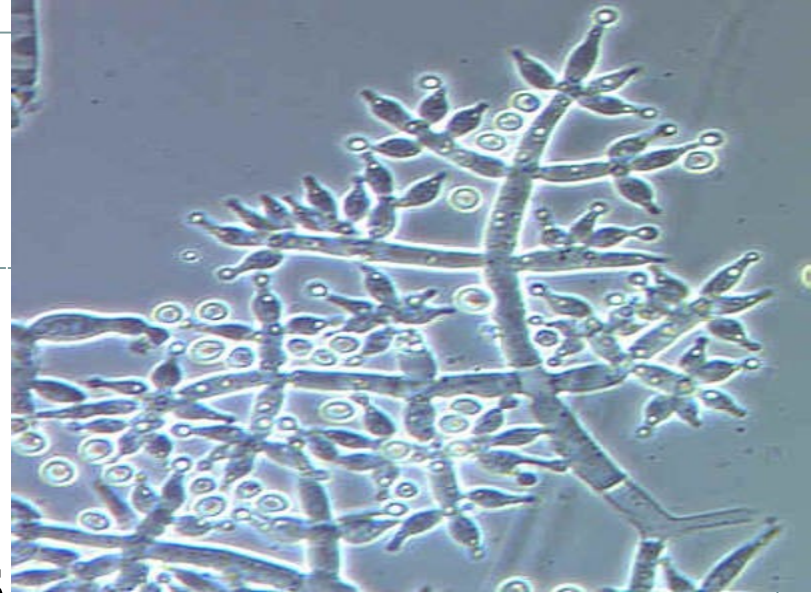
Frankia and actinorrhizal plant Symbiosis



Major groups of soil microorganisms :



- Fungi:
 - the major component of microbial biomass in soils
 - major participants in decomposition of organic matter
 - hyphal growth helps bind soil particles into stable aggregates.



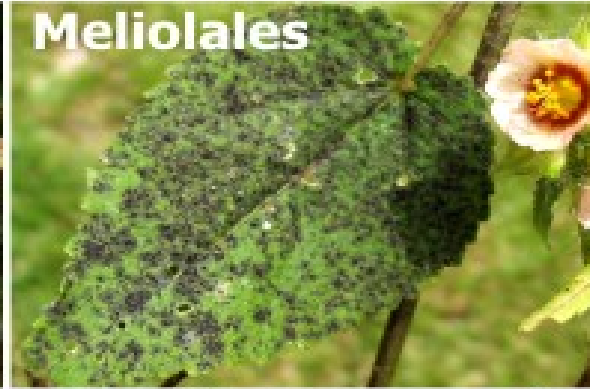
Tropical plant parasitic microfungi belonging to different systematic groups of ASCOMYCOTA

Plant
communities are
closely
associated with
fungal and
bacterial
communities in
their soil

Taphrinales



Meliolales



Hypocreales



Phyllachorales



Erysiphales



Helotiales



Rhytismatales



Mycosphaerellales



Microthyriales



Major groups of soil microorganisms :



Fungi:

- Some associate with plant roots: major plant pathogens,
- Beneficial symbionts increase nutrient uptake and decrease disease incidence

Major groups of soil microorganisms :



Cyanobacteria and algae (green algae, diatoms):

- Photoautotrophs, form surface algal crusts important in H₂O retention
- Some cyanobacteria carry out free-living and symbiotic N₂-fixation

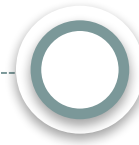
Major groups of soil microorganisms :



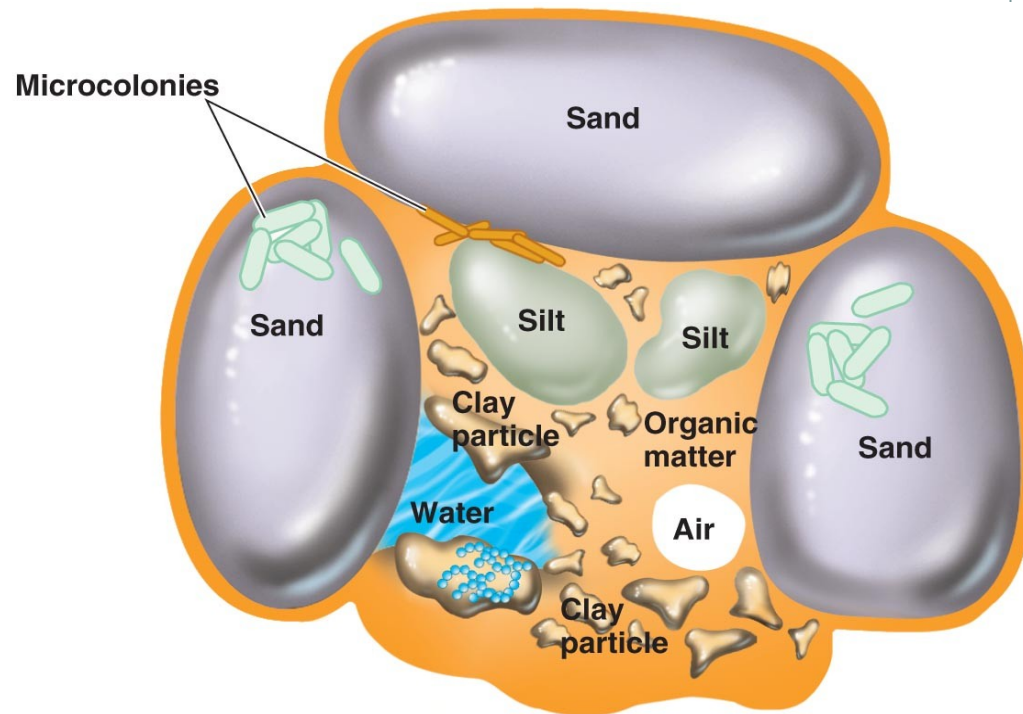
Viruses:

- Numerically abundant, ecology not well defined
- bacteriophage
- Persistence and migration of human enteroviruses pose serious health issues with land disposal of sewage and fecal wastes

Terrestrial Environments



- Most microbial growth takes place on the surfaces of soil particles
- Soil aggregates can contain many different microenvironments supporting the growth of several types of microbes



Rhizosphere: the Place to be in Soil

What is the Rhizosphere

Rhizo= Rhiza= root

Sphere= field or area of influence ,
physical location around the root

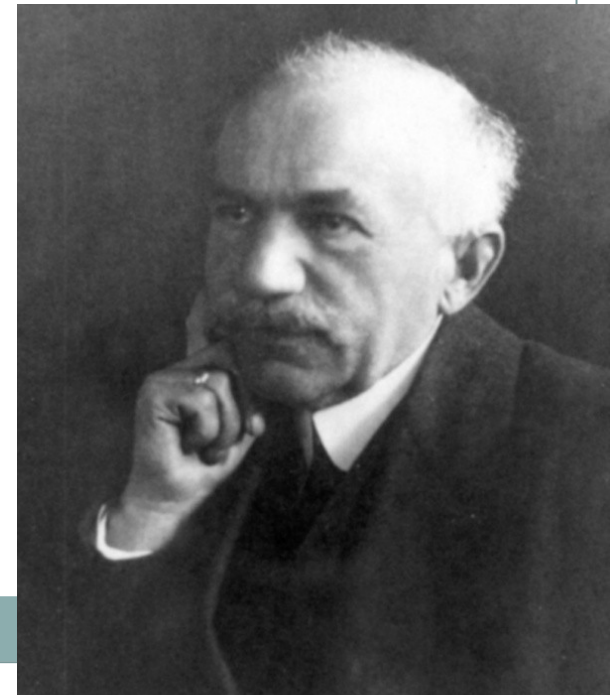
Introduced by Hiltner in 1904

the region surround plant roots

influence by root exudates

1-2 mm from root

variable region where the root
production stimulate microbial
population and activity



Rhizosphere



Comprised of different zones

Endorhizosphere

multi-layered microenvironment, which includes a mucoid layer on the root surface, the epidermal layer of the root tissue including the root hairs and the cortical cells

Ectorhizosphere

rhizosphere soil



Rhizosphere

It's plant's root surface (external surface)
where many microorganism adhere to it.
1μm thick.

hard to distinguish between Rhizo sphere/
plane.

NonRhizosphere

Zone that not penetrated by plant root .
it has less microbial activity.

Rhizosphere vs NonRhizosphere

In the Rhizosphere:

More enhanced microbial growth

More soluble C, nutrients



Distance (μm)	Morphologically distinct microbe type	Numbers (x 10 ⁻⁹)
0-1	8	120
0-5	11	96
5-10	6	41
10-15	3	34
15-20	3	13

Organic material in rhizosphere

- **Exudates**

- low molecular weight compounds released from plant cells in a non-metabolic manner (leakage)

- **Secretions**

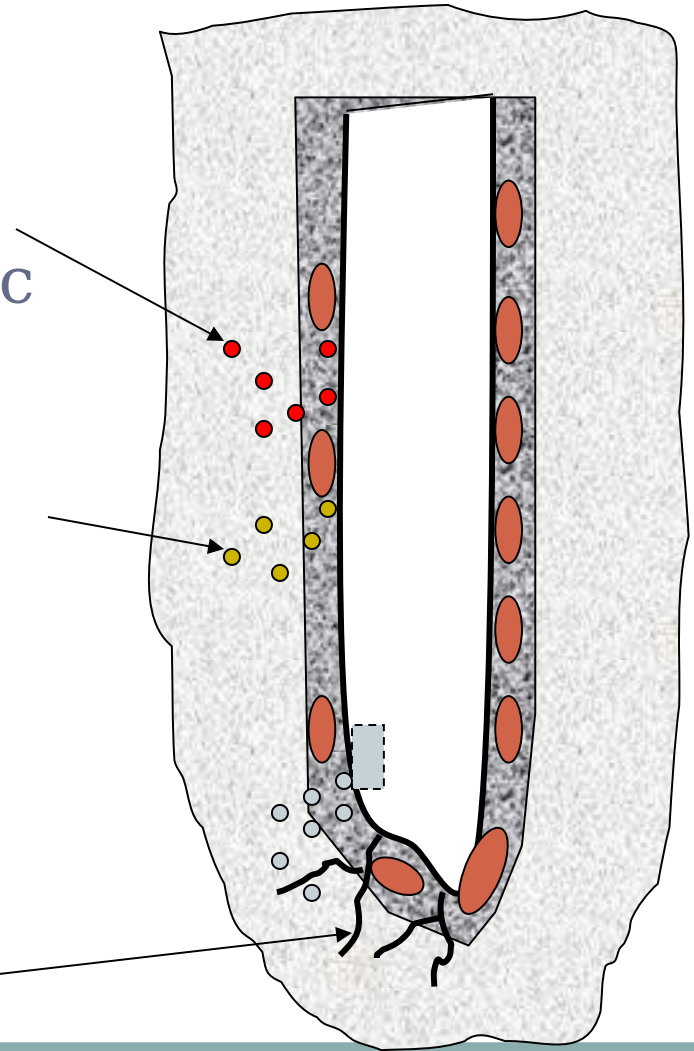
- compounds metabolically released from plant cells

- **Lysates**

- compounds released from cells during autolysis

- **Plant mucilage**

- plant polysaccharides



Densities of microorganisms in rhizosphere

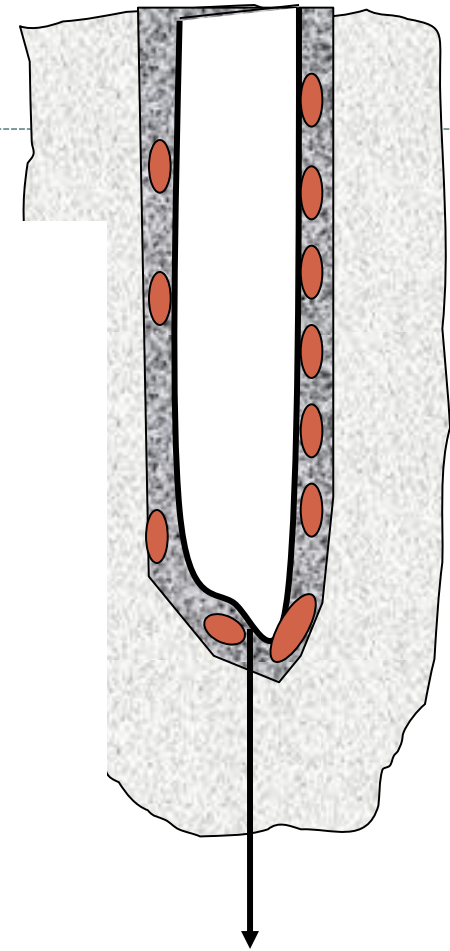


TABLE 18.1 Numbers of Microorganisms in the Rhizosphere (R) of Wheat (*Triticum aestivum*) and Nonrhizosphere Soil (S) and Their Resultant R/S Ratio

Microorganisms	Rhizosphere	Nonrhizosphere	R/S ratio
	CFU ^a g soil	CFU ^a g soil	
Bacteria	120×10^7	5×10^7	24.0
Fungi	12×10^5	1×10^5	12.0
Protozoa	2.4×10^3	1×10^3	2.4
Ammonifiers	500×10^6	4×10^6	125.0
Denitrifiers	1260×10^5	1×10^5	1260.0

From Rouatt *et al.* (1960).

^a CFU, colony-forming units.



Decreasing
microbial
diversity

Mucilage



Insoluble organic compounds of four different origins.

- mucilage secreted by Golgi organelles in the root cap cells.
- hydrolysates of the polysaccharides of the primary cell wall between epidermal cells of the primary wall and sloughed root cap call.
- mucilage secreted by epidermal cells and root hairs.
- mucilage produced by bacterial degradation of dead epidermal cells.

Root exudates

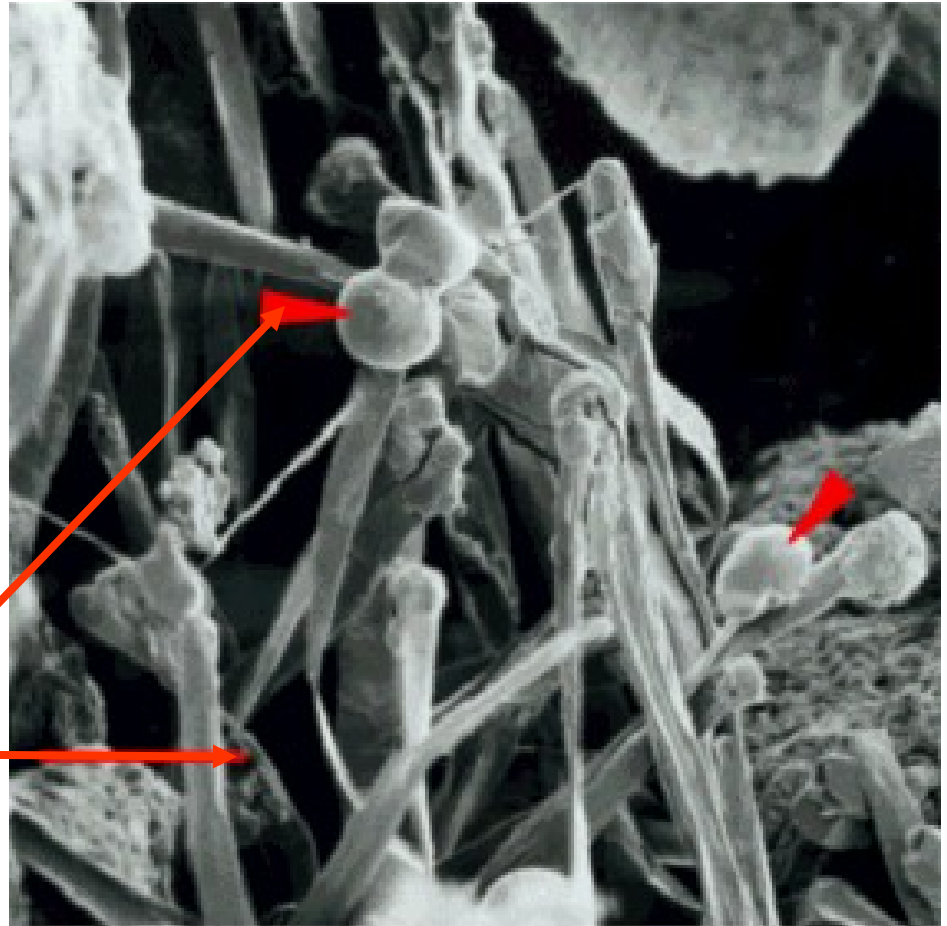


Low molecular weight compounds.

Act as a messengers

Energy for microbes

Root exudates as drops on root surface



Root exudates function

Defend the rhizosphere and root against pathogenic microorganism

Attract greater number of microorganism

Keep the soil around the root moist

Obtain nutrients

Change the chemical properties of the soil around the root

Inhibit the growth of competing plant species

Enzymes in the rhizosphere

Enzymes activity greater in rhizosphere than in bulk soil

Wide range of enzymes (plant- microorganism)

- Oxidoreductases (transfer of electrons from one molecule to another)
- Hydrolases (nutrient mineralization, protein breakdown)
- Lyases (degrades cell wall and membrane of bacteria and fungi)
- Transferases (transfer a functional group from one molecule to another)

Enzymes catalyze the breakdown of

- organic material (cellulases, dehydrogenases)
- fertilizers (ureases)
- organic nutrients (phosphatases, sulfatases)

Some function of Bacteria in rhizosphere

Bacteria N₂ fixation

Rhizobium, *Bradyrhizobium* with legumes

Azospirillum on roots surface of grass

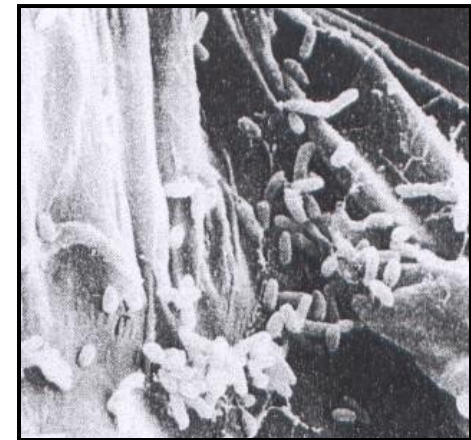
Frankia with nonleguminous angio



Rhizobium



Frankia



Azospirillum

Some function of Bacteria in rhizosphere



Bacteria has role in P cycle in the rhizosphere

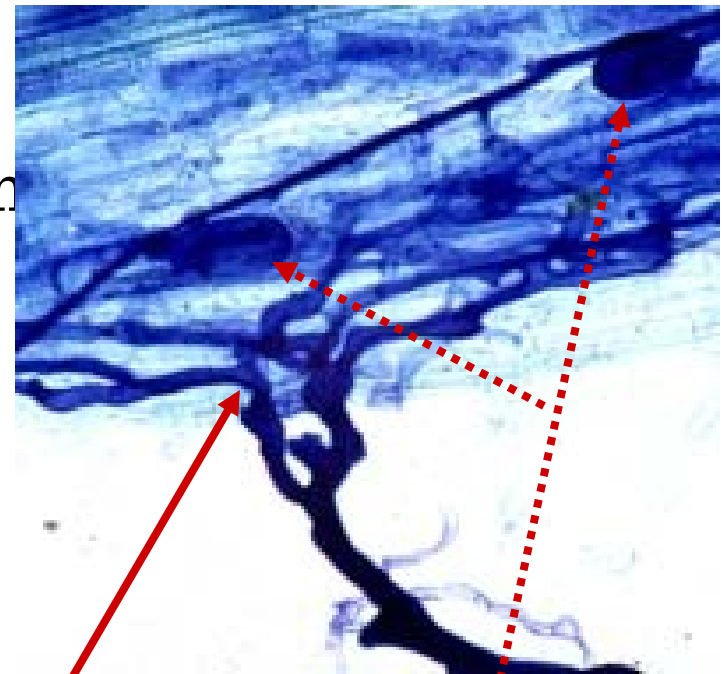
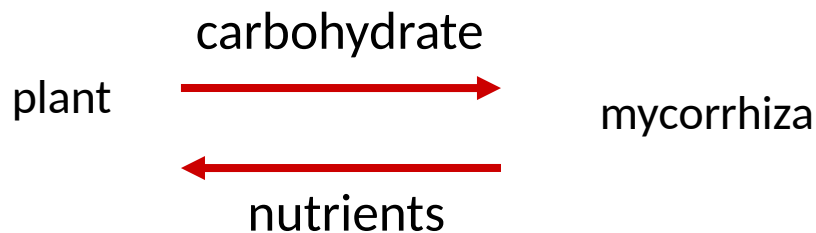
Solubilization of inorganic phosphorus (P dissolving bacteria) *Pseudomonas*, *Bacillus*

Reduce pH by secret organic acids (formic, acetic, lactic acid)

Mycorrhizal fungi



- Beneficial fungi associate with plant root
- ~ 90 % of plant count on mycorrhizal fungi root system
- nutrient exchange



hyphae

Vesicular Arbuscular

Microorganism in rhizosphere



- **Nematodes**
- Up to 83% soil nematodes are root feeders (herbivores)
- Nematode activity stimulate by substances in Rhizosphere (CO_2 , ethylene- amino acids- salts- mucigel)



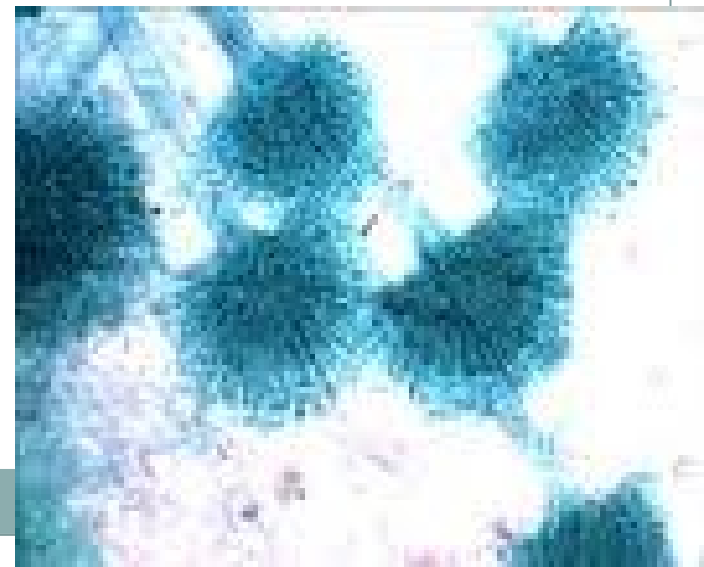
Mycorrhizal fungi, especially ectomycorrhizal (sheathing mycorrhizal) and ericoid mycorrhizal fungi, which were recognized recently as potentially important for mineral weathering (Lndeweert *et al.* 2001).

Ectomycorrhizae on roots of southern yellow pine

Roots of saplings (aged 10-15) in a long-term study of forest succession



Saprotrophic fungi, *Aspergillus niger* which use relatively simple carbohydrate sources and quickly proliferate under favorable conditions.





Saprotrophic organisms that use the lignocellulose complex.

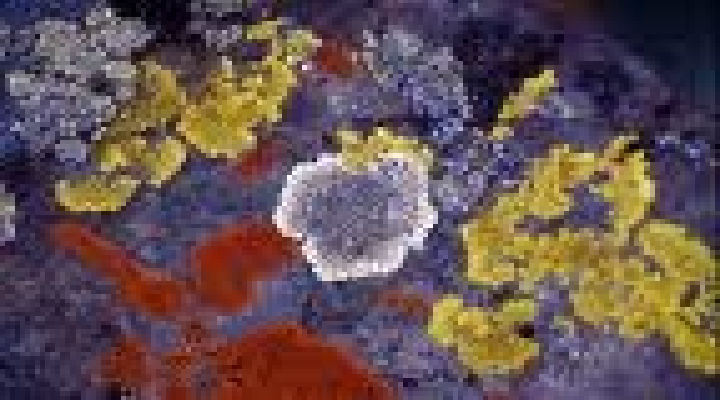
White rot and brown rot fungi produce large amounts of organic anions (especially oxalate) that they use for generating hydrogen peroxide and for the production of Mn chelates that can oxidize a range of phenolic compounds may therefore be potential weathering agents.



How do fungi weather rocks and minerals?

Fungi can penetrate solid materials using both physical and chemical tools. The mycelial growth form allows fungi to exploit weak spots on the rock or grain surface.





Lichen compounds. "Lichen acids", produced by the fungal

partner in the lichen symbiosis, have for a long time been recognized as the major weathering components of lichens growing on rocks.

Furthermore, lichen compounds are usually concentrated in the upper thallus, at the fungal-algal or cyanobacterial interface, not at the mineral interface.

In the field, metal-lichen-acid complexes have very rarely been reported, suggesting they play a limited role in weathering.



Decomposition and soil organic matter





Plant Litter Decomposition

phases:

Leaching by water. Removes soluble materials (nutrients, simple C

Fragmentation by soil animals.

Chemical alteration. Changes chemical composition of detritus.

organic matter molecules

Compounds are decomposed at different rates,
new compounds are synthesized

Microbial activity controlled by:

Environment: soil T, H₂O, soil O₂

Substrate: chemical makeup of organic matter

Rate of microbial activity determines rate of C
loss, nutrient release etc.

Rate of activity measured as organic matter mass
loss.

Fungi

Accounts for most aerobic decomposition

- 60-90% of microbial biomass in forests
- About half of microbial biomass in grassland

Broad enzymatic capability

- Cell walls (lignin, cellulose, hemicellulose)
- Cell contents (proteins, sugars, lipids)

Can transport metabolites through hyphae

- Surface litter (import nitrogen from soil)
- Wood degraders (import nitrogen from soil)

- **Bacteria**

Grow rapidly on labile substrates

Some function anaerobically

Dependent on substrates that diffuse. Diffusion gradient caused by

- Production of soluble substrates (enzymes)

- Uptake of substrates by bacterium

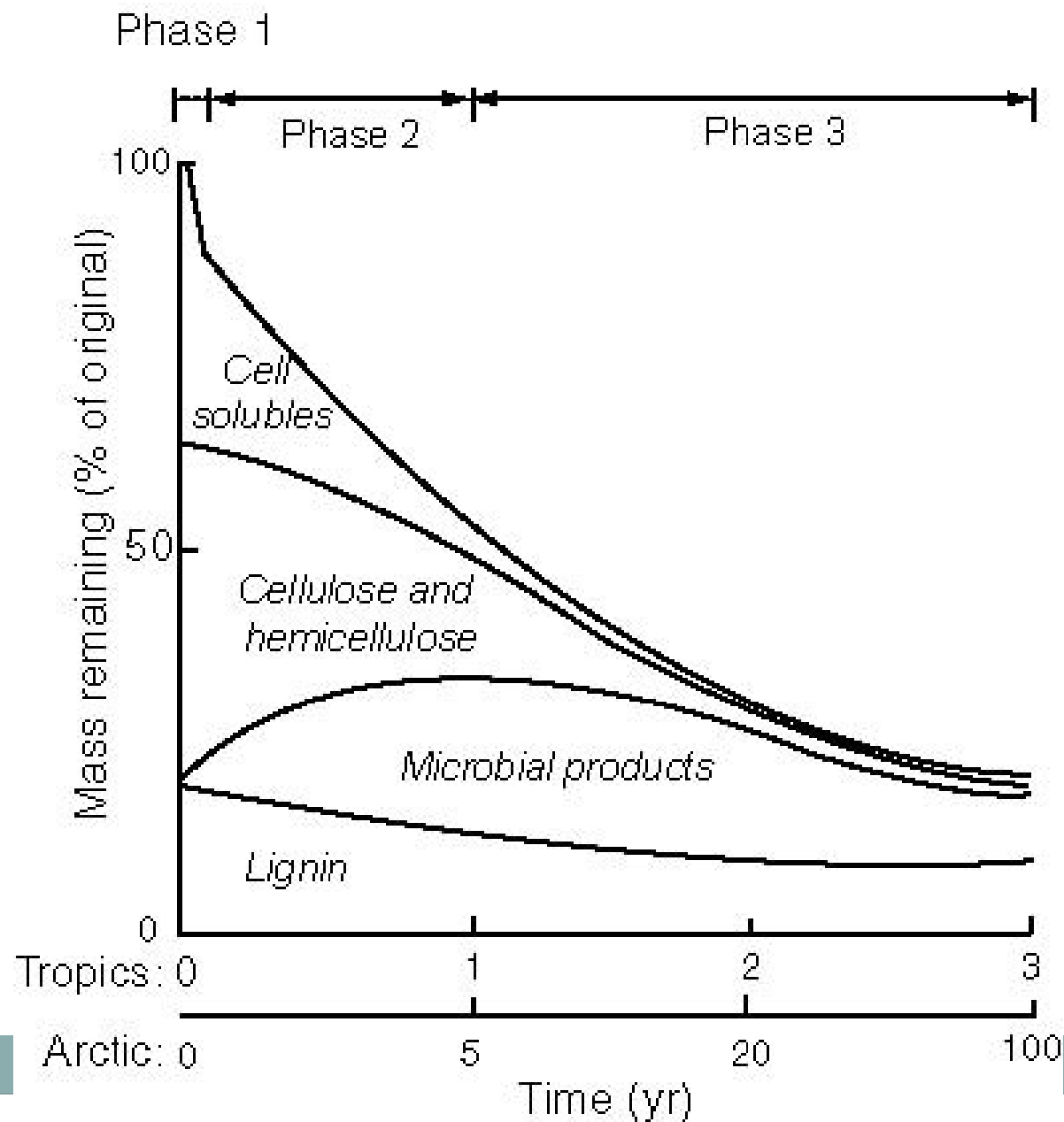
Spatial specialists

- Rhizosphere, macropores, interior of aggregates

- form biofilms on particle surfaces

Chemical specialists

- Different bacteria produce different enzymes



Organic matter in the soil

Organic matter is anything that contains carbon compounds that were formed by living organisms. Three main components are

Dead forms of organic material - mostly dead plant parts (85%)

living parts of plants - mostly roots (10%)

living microbes and soil animals

Partly decayed organic matter is called humus

continuum of organic matter decomposition from fresh
litter to well decomposed humus

Chemical Separation

Fulvic acids- soluble in acid, base

Humic acids- soluble in base, not acid

Humins - not soluble in base or acid

Major steps in the degradation of organic matter

Involves 4 major steps

Breakdown of compounds that are easy to decompose (e.g. sugars, starches and proteins)

Breakdown of compounds that take several years to decompose (cellulose and lignin)

Breakdown of compounds that take 10 years (e.g. waxes and phenols)

Compounds that takes 100-1000's of years (e.g. humus like substances, which are very complex)

Microbial degradation of organic matter

Involves complex processes. Chemical alteration of organic matter, physical fragmentation and release of mineral nutrients.

The dead organic matter is colonized by microbes and degraded with help of microbial enzymes.

Macromolecules are broken down into simpler units and further degraded into constituent elements.

Microbial degradation of organic matter..

Why do stems and leaves degrade at different rates?

Because they are made up of different kinds of molecules and require different enzymes.

E.g. Cellulose, predominant in leafy tissues are easily degradable than lignin, major component of wood.

Mineralization of organic matter

Mineralisation is the biological process where organic compounds in organic matter are chemically converted by the microorganisms in soil to simpler compounds, other organic compounds or mineralised nutrients.

Examples of tropical and temperate region plant-soil systems

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Tropical soil

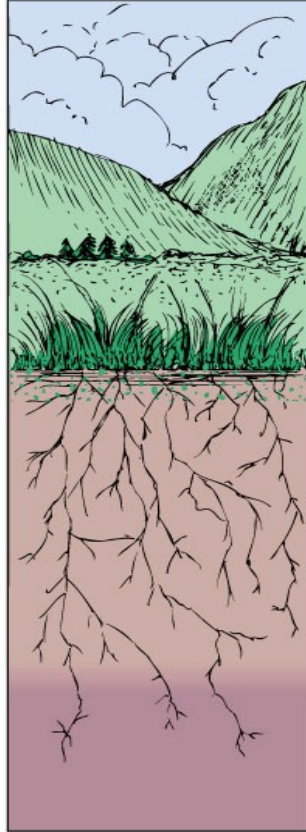


Rapid litter decomposition

Low organic matter soil

Roots penetrate into the litter layer to absorb nutrients, which minimizes losses by leaching

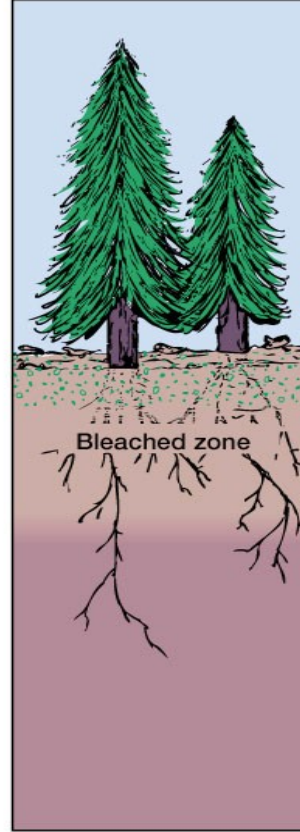
Temperate grassland



Litter can accumulate on surface

Deep root penetration and organic matter accumulation

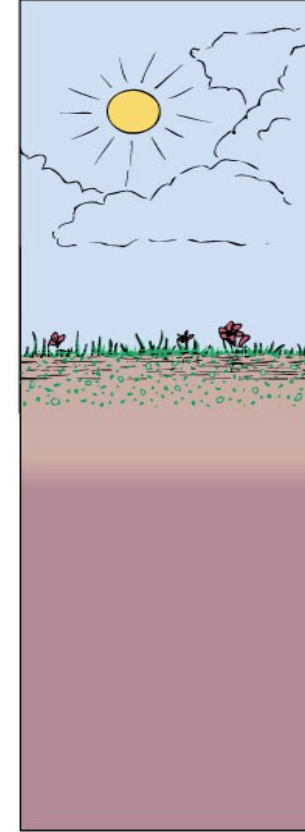
Temperate coniferous



Moist litter layer with lower oxygen levels

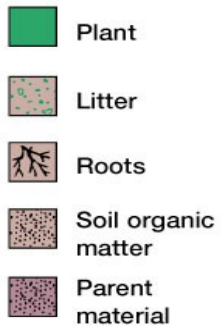
Acidic products accumulate and leach into the underlying soil, resulting in the formation of a bleached zone

Bog soil



Litter accumulation due to lower soil oxygen levels and decreased degradation

Formation of peat



Humification

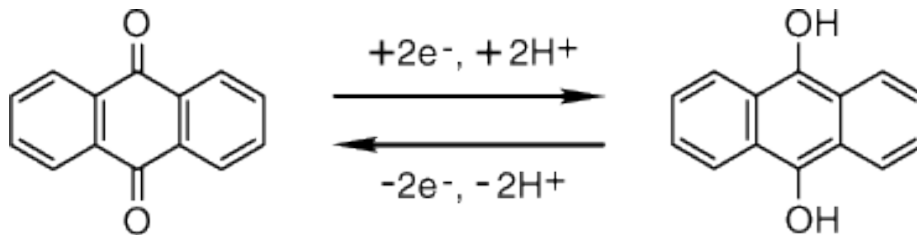
Selective preservation -- plant waxes, cutins, suberin, lignin, chitin, microbial cell walls are hard to decompose & become enriched in soil

Microbial transformations -- enzymatic breakdown by microbes produces low molecular weight soluble compounds such as amino acids.

Polyphenol formation --phenols in soil are derived from lignin breakdown, soil microbial synthesis, and by plants as defensive compounds

Humification

4. **Quinone formation** -- conversion of polyphenols to reactive quinones as a result of extracellular enzymes



5. **Abiotic condensation** -- quinones react spontaneously with many soil compounds such as amino acids to form humus

What stimulates soil organic matter decomposition?

- Usual suspects (changes in climate, nutrient availability, aeration)
- Mechanical mixing/plowing
- Plant species change
- C additions “priming the pump”...
- Soil weathering...?

SOIL FORMATION AND AGGREGATION

Where are the microorganisms located in the soil?

Usually the top 2-3 cms.

Commonly found close to root surfaces, in dead roots, on soil particles or amongst aggregates of soil particles.

Soils that are clayey have many bacteria because these soils have lot of small pores.

Sandy soil is less suitable habitat.

Soil aggregates and soil pores

What are soil aggregates and soil pores?

Soil aggregates: Clumps of soil particles that are held together by moist clay, organic matter, gums (from bacteria and fungi) and by fungal hyphae.

Vary in size from 2 microns to 2 mm.

Soil pores: spaces between soil particles and soil aggregates.

They can be full of air or have water in them.

Very essential for storing air, water, microbes, nutrients and organic matter.

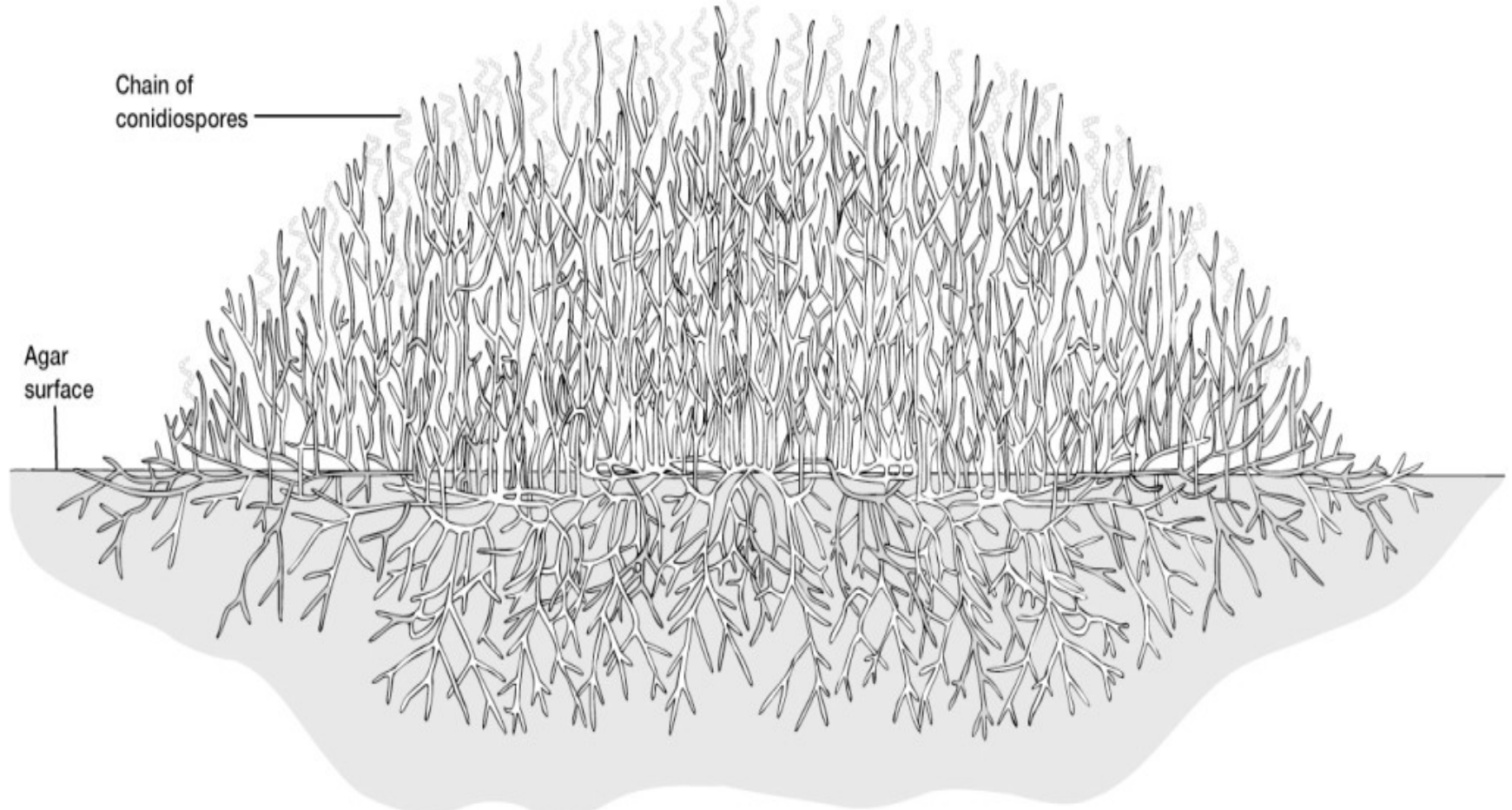
Bacterial/ Fungal role in the formation of soil aggregate

Two ways

- Using bacterial polysaccharide as a gum
- Through electrostatic charge between bacteria and clay surfaces
- Fungi help form soil aggregates when fungal hyphae enmesh and cross link soil particles.
- Mycorrhizae are very important in this process.

ACTINOMYCETES MORPHOLOGY

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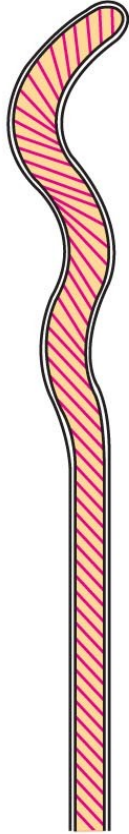


An actinomycete colony growing on agar - note the subterranean and aerial hyphae....

Spore Formation in *Streptomyces*



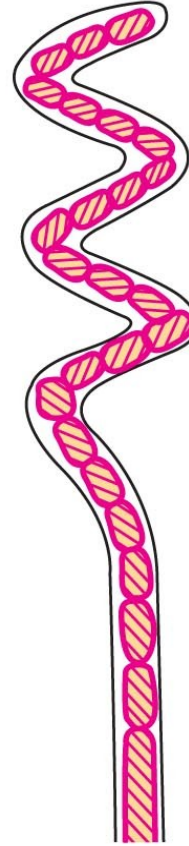
**Growth
phase**



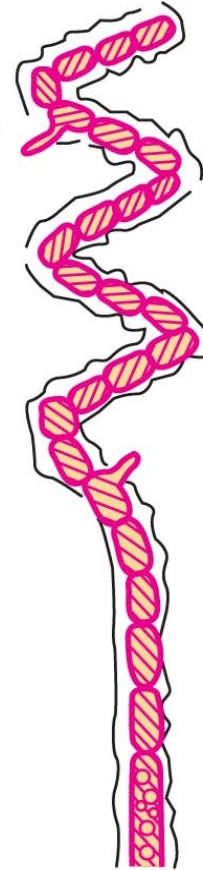
**Tip
curls**



**Partitioning
of tip**



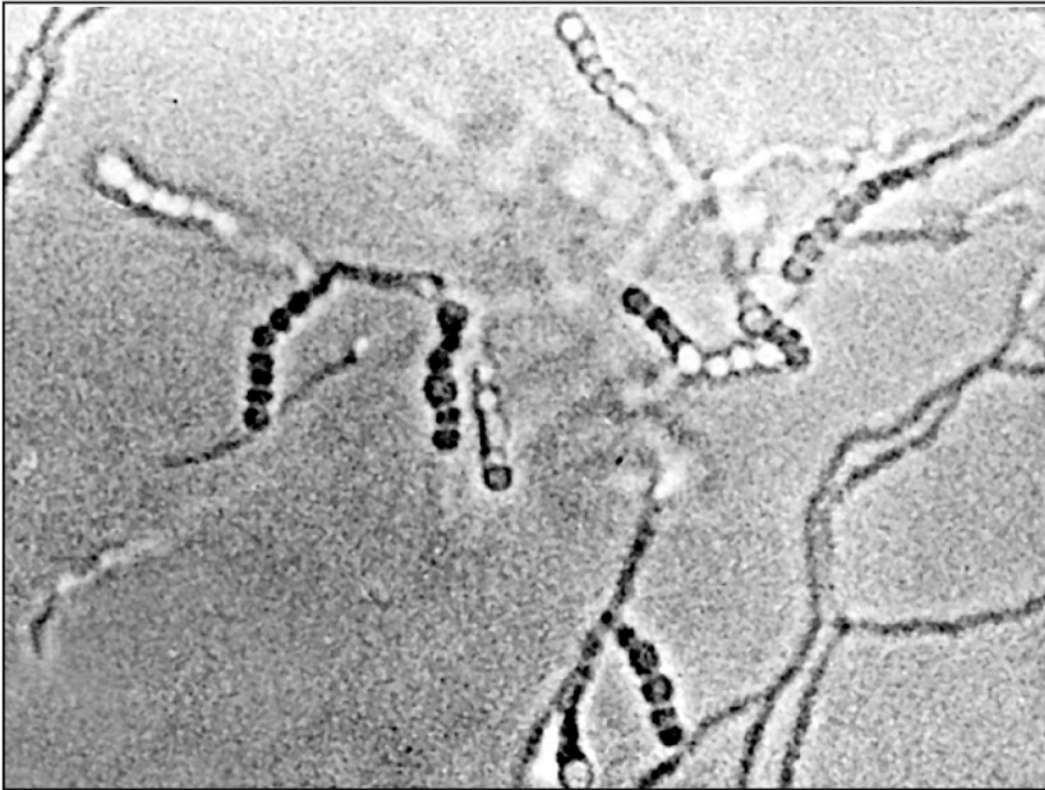
**Cell walls
thicken
and constrict**



**Spores
mature**

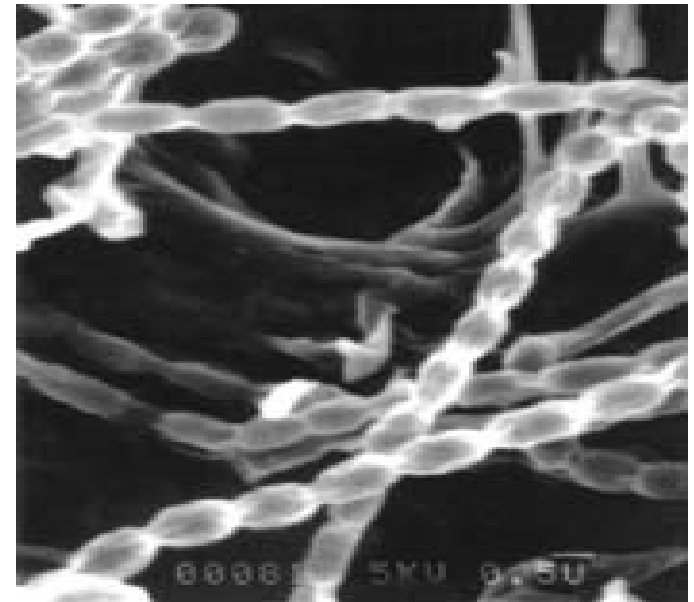
ACTINOMYCETES REPRODUCTION

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Conidia of *Streptomyces*

5 μ m



High Magnification of
actinomycete spores

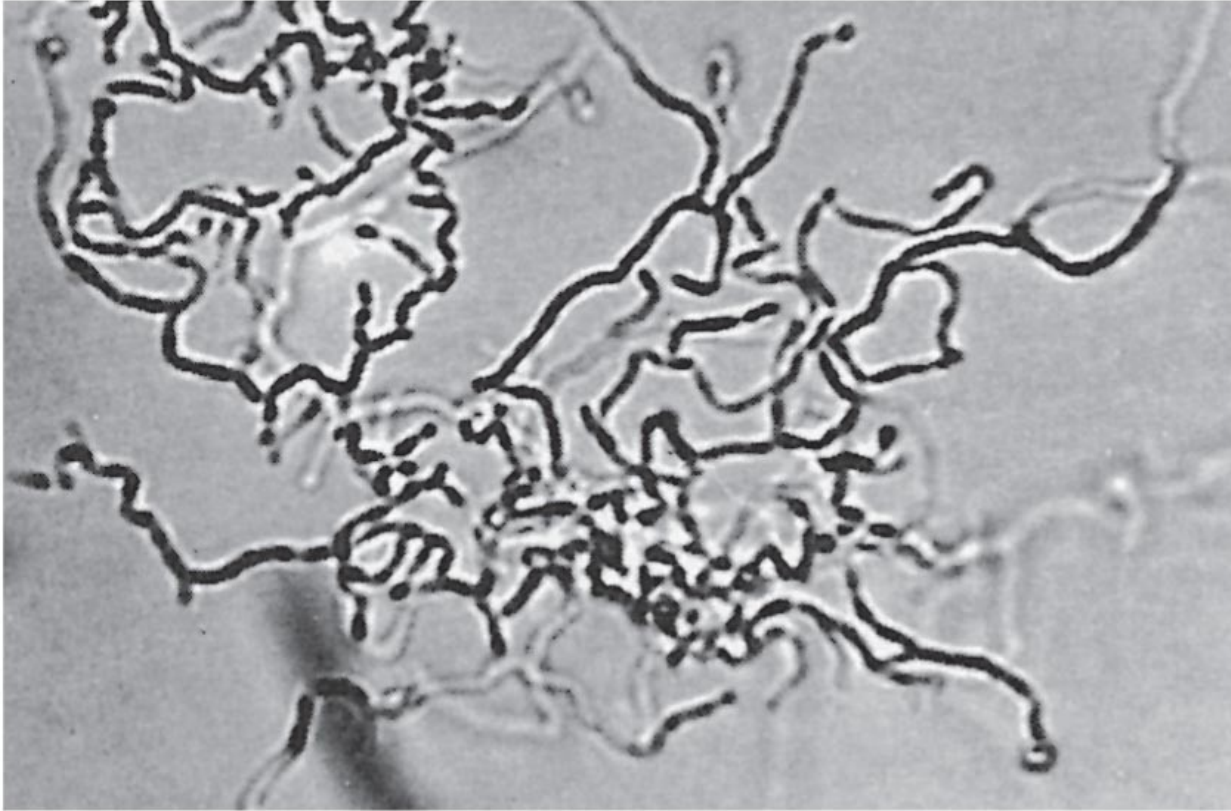
ACTINOMYCETES REPRODUCTION

Actinomycetes (and fungi) reproduce via spores
hyphal growth is followed by fragmentation
and release of spores

conidia – spores produced asexually on aerial
filaments called sporophores

shape and arrangement of sporophores help
identification of actinomycete taxa

Nocardia



Hubert and Mary P. Lechevalier

Figure 16.18

Spore-Bearing Structures of Actinomycetes



Peter Hirsch

(a) *Streptomyces*
sp.

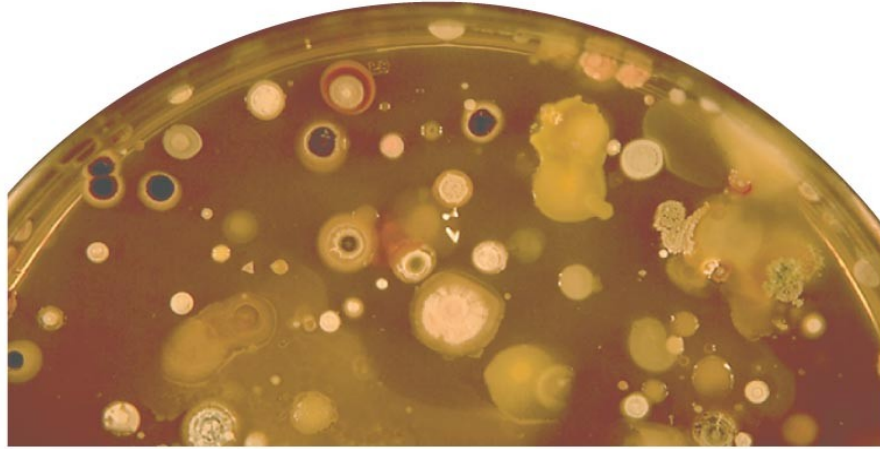


Hubert and Mary P. Lechevallier

(b) *Streptomyces*
sp.

Figure 16.19

Colonies of *Streptomyces*



M. T. Madigan

(a)



David A. Hopwood

(b)

Figure 16.22

ACTINOMYCETES ECOLOGY



predominantly soil bacteria
good at degrading recalcitrant compounds
such as chitin & cellulose
often active at higher pH (contrast to fungi who
may dominate at lower pH)
give soil the “earthy” smell - geosmin

Examples



Genus *Frankia* –

large genus that all form nodules on certain non-legume trees

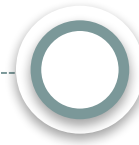
fixes nitrogen from the air and converts into forms useable by plant host

involved in relationship with plant and mycorrhizal fungus. e.g.

Myrica in the tropics (invasive in Hawaii)

Alnus in temperate climates (early successional)

Frankia nodules on *Ceanothus* roots



Other antibiotics produced by *Streptomyces* spp

Antibiotic:

Spectinomycin
gonorrhoeae

affects:

M. tuberculosis, Neisseria

Neomycin

Broad spectrum,

Tetracyclines

Broad spectrum, Chlamydias,
Rickettsias etc.

Nystatin

Fungi, esp. *Candida*

Erythromycin

Gram + Bacteria and *Legionella*

Chloramphenicol

Broad spectrum, typhoid fever

Recommended Reading-General Microbiology



- GJ Tortora, Christine L and Funke BR (2007). Microbiology: an introduction, 9th ed. Pearson Education, New Delhi.